

10 Chemical reaction II

(Electron and Proton transfer reactions)

1. Electron transfer reactions

2012-5.

- a. Define “electroplating”. (2marks)
- b. Iron(Fe) displaces copper(Cu) from copper sulphate solution(CuSO_4).
- (i) Write down a balanced chemical equation for the reaction. (3marks)
- (ii) What is the reducing agent in the reaction? (1mark)
- (iii) Give a reason for the answer in 5.b.(ii). (1mark)
- c. What is the difference between “oxidation” and “reduction”? (2marks)

2011-3.

- d.
- (i) Mention any two ways of preventing corrosion in metals. (2 marks)
- (ii) Calculate the oxidation number of nitrogen (N) in a nitrate ion (NO_3^-) given that the oxidation number of oxygen (O) is -2 . (3 marks)

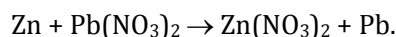
2010-2.

- a. State any two advantages of electroplating a metal.
- b. Below is part of a displacement series of metals.

Calcium (Ca)		Increasing reactivity
Magnesium (Mg)		
Zinc (Zn)		
Iron (Fe)		
Copper (Cu)		

- (i) Which metal would displace all other metals from their solution? (1 mark)
- (ii) Give a reason for the answer in 2.b.(i). (1mark)
- (iii) What would happen if a piece of magnesium metal was placed in copper sulphate solution? (2 marks)
- d. Define “oxidation” in terms of oxygen content in a substance. (1 mark)

e. The chemical equation below shows displacement reaction between zinc (Zn) and lead nitrate ($\text{Pb}(\text{NO}_3)_2$).



Name the reducing and oxidising agents in the reaction.

Reducing agent: (1 mark)

Oxidising agent: (1 mark)

f. **Figure 2** is a diagram showing the set-up of an experiment to investigate conditions for rusting of iron.

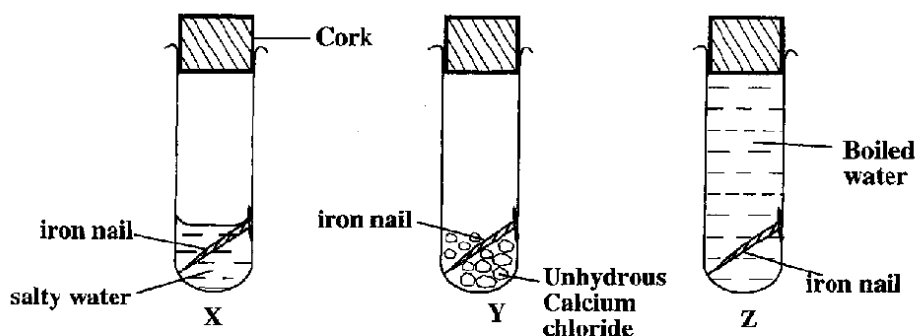


Figure 2

(i) Why was unhydrous calcium chloride used in tube Y? (1 mark)

(ii) Why was water in tube Z boiled? (1 mark)

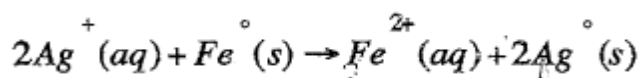
(iii) State any two condition necessary for rusting of iron. (2 marks)

2010-7.

b. With the aid of a labelled diagram, explain how a silver spoon could be electroplated using copper. In the explanation, include the half equation for the reaction at the cathode. (11 marks)

2009-5.

e. Silver ions ($\text{Ag}^+(\text{aq})$) react with iron (Fe) according to the following equation:



(i) what is the meaning of (2+) of $\text{Fe}^{2+}(\text{aq})$? (1mark)

(ii) What is the oxidation number of silver before reaction? (1mark)

(iii) Which substance has been reduced? (1mark)

(iv) Give a reason for the answer in 5.e.(iii). (1mark)

2009-8.

a. What is an “electrolyte”? (2marks)

b, With the aid of a well labeled diagram, describe an experiment that can be carried out to compare the electrical conductivity of potassium nitrate solution and potassium chloride solution. (13marks)

2008-7.

b. What is the difference between “oxidation” and “reduction” in terms of electron transfer? (1 mark)

c. Explain how each of the following prevents resting of iron.

(i) painting (4 marks)

(ii) galvanizing (5 marks)

2007-3.

a. Ammonia is an example of a strong base.

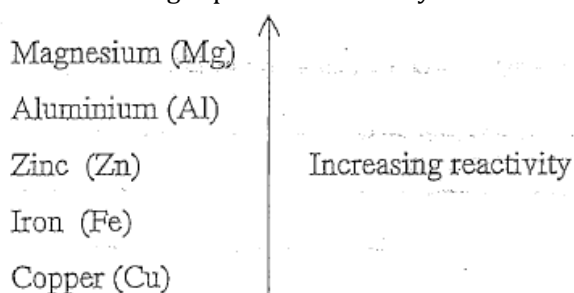
(i) What is a “strong base”? (1mark)

(ii) Write a chemical equation to show the ionization of ammonia in water. (3marks)

(iii) Identify one conjugate acid-base pair from the equation in 3.a.(ii). (1mark)

2007-6.

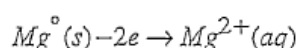
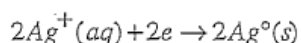
e. The following is part of a reactivity series.



(i) Which two elements will displace zinc (Zn) from its oxide? (2marks)

(ii) Give a reason for the answer to 6.e.(i). (1mark)

f. The following are half equations for the reaction between magnesium (Mg) and silver nitrate (AgNO_3):

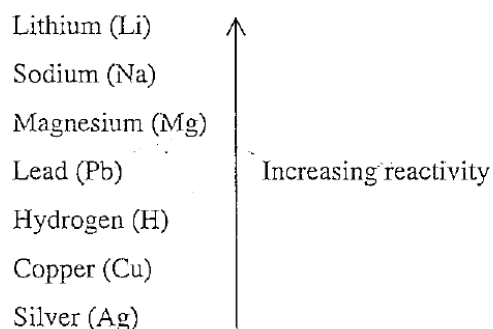


(i) Write a full chemical equation for the reaction. (2marks)

(ii) Name the reducing and oxidizing agents in 6.f.(i). (2marks)

2006-5.

d. the following is part of an activity series.



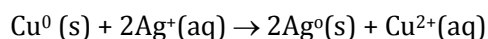
- (i) State whether copper (Cu) will react with a solution of magnesium sulphate (Mg SO₄) (1mark)
 - (ii) Explain the answer to 5.d.(i). (2marks)
 - (iii) Which element is the most electropositive in the activity series? (1mark)
 - (iv) Give a reason for the answer to 5.d.(iii). (2marks)
- e.
- (i) Write half equations for the reaction between silver nitrate (AgNO₃) and Sodium (Na). (2marks)
 - (ii) Name the reducing and oxidizing agents in 5.e.(i). (2marks)

2005-7.

- a. Draw a labeled diagram of the apparatus that would be used to electroplate an iron nail with copper using copper chloride as an electrolyte. (4marks)
- b. Explain what happens during the process of electroplating of the iron nail in (a). Support the explanation with relevant chemical equations. (11marks)

2003-3.

- b. Copper (Cu) reacts with silver ions (Ag⁺) according to the following chemical equation:



- (i) What is the meaning of the zero sign(0) in Cu⁰ (s)? (1mark)
- (ii) Pick out the oxidizing agent and the reducing agent from the equation.
 - Oxidizing agent : (1mark)
 - Reducing agent : (1mark)
- (iii) Write the two half equations for the reaction. (4marks)

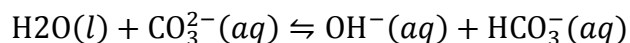
2. Proton transfer reactions

2011-3.

- b.
 - (i) Define an “acid” according to Lowry/Bronsted theory. (1 mark)
 - (ii) Describe how a hydronium ion (H₃O⁺) is formed. (2 marks)

2010-2.

- c. Water reacts with the carbonate ion according to the following chemical equation.



Give one conjugate acid-base pair in the reaction. (3 marks)

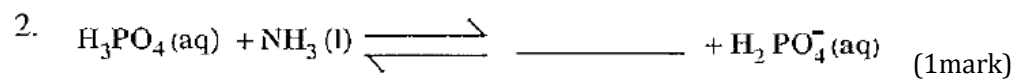
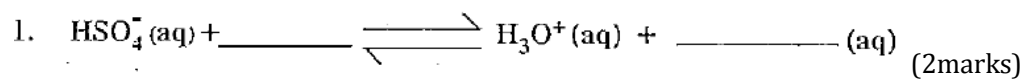
2006-5.

- a.
 - (i) Name the ion responsible for the acidic properties of a substance. (1mark)
 - (ii) Why is carbonic acid a weak acid while hydrochloric acid a strong acid? (2marks)

c.

(i) What does the symbol $\rightleftharpoons$ mean in a chemical equation? (1mark)

(ii) Complete the following chemical equations:



2005-3.

a. Define "acid" according to Bronsted Lowry theory. (1mark)